

Citywide Model Comparison - Geospatial Data: 2022 - 2025 Sales

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Model I

Table 1: Model Metrics: LN-SPPSF Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	19756	19732	23	0.4026	0.1621	0.1611	0.5351	165.9671	31383	31580
+ Geospatial	19748	19723	24	0.4042	0.1633	0.1623	0.5347	160.4483	31345	31550
% Difference	-0.04%	-0.05%	4.35%	0.39%	0.77%	0.75%	-0.07%	-3.33%	-0.12%	-0.09%

Table 2: Ratio Metrics: LN-SPPSF Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9884	1.1561	0.8753	1.3209	48.5316	-0.9258	13.50%	27.84%	67.83%
+ Geospatial	0.9871	1.1562	0.8756	1.3204	48.6087	-0.9249	13.50%	27.83%	67.92%
% Difference	-0.13%	0.00%	0.04%	-0.04%	0.16%	0.10%	-0.03%	-0.05%	0.13%

Model II

Table 3: Model Metrics: LN-PRICE (No ECF) Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	19756	19729	26	0.6092	0.3712	0.3703	0.5328	447.9058	31218	31438
+ Geospatial	19748	19718	29	0.6105	0.3728	0.3718	0.5323	404.0598	31166	31411
% Difference	-0.04%	-0.06%	11.54%	0.21%	0.42%	0.40%	-0.11%	-9.79%	-0.17%	-0.09%

Table 4: Ratio Metrics: LN-PRICE (No ECF) Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9869	1.1542	0.8765	1.3168	48.2366	-0.4657	13.67%	27.87%	68.28%
+ Geospatial	0.9860	1.1541	0.8771	1.3158	48.2846	-0.4648	13.74%	27.96%	68.29%
% Difference	-0.10%	-0.01%	0.07%	-0.07%	0.10%	0.21%	0.48%	0.31%	0.01%

Model III

Table 5: Model Metrics: LN-PRICE Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	19756	19569	186	0.7158	0.5124	0.5077	0.4711	110.5488	26514	27997
+ Geospatial	19748	19718	29	0.6105	0.3728	0.3718	0.5323	404.0598	31166	31411
% Difference	-0.04%	0.76%	-84.41%	-14.71%	-27.25%	-26.77%	12.98%	265.50%	17.55%	12.19%

Table 6: Ratio Metrics: LN-PRICE Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9849	1.1155	0.9059	1.2314	41.5992	-0.3079	15.08%	30.71%	73.52%
+ Geospatial	0.9860	1.1541	0.8771	1.3158	48.2846	-0.4648	13.74%	27.96%	68.29%
% Difference	0.11%	3.46%	-3.18%	6.86%	16.07%	-50.95%	-8.92%	-8.98%	-7.12%

Model IV

Table 7: Model Metrics: LN-PRICE (Inliers) Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	18400	18212	187	0.7591	0.5762	0.5719	0.4073	132.4159	19355	20833
+ Geospatial	18421	18393	27	0.6713	0.4506	0.4498	0.4583	558.8244	23562	23789
% Difference	0.11%	0.99%	-85.56%	-11.56%	-21.79%	-21.34%	12.51%	322.02%	21.74%	14.19%

Table 8: Ratio Metrics: LN-PRICE (Inliers) Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9832	1.0856	0.9287	1.1689	36.3722	-0.2367	16.40%	33.43%	78.03%
+ Geospatial	0.9922	1.1102	0.9092	1.2211	41.0284	-0.3554	14.97%	30.16%	72.61%
% Difference	0.92%	2.27%	-2.10%	4.47%	12.80%	-50.16%	-8.75%	-9.81%	-6.95%

Model V

Table 9: Model Metrics: LN-TASP Model Comparison

Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	18400	18215	184	0.7535	0.5678	0.5634	0.4074	130.0333	19353	20808
+ Geospatial	18421	18396	24	0.6628	0.4393	0.4386	0.4583	600.5914	23559	23762
% Difference	0.11%	0.99%	-86.96%	-12.04%	-22.62%	-22.15%	12.51%	361.88%	21.73%	14.20%

Table 10: Ratio Metrics: LN-TASP Model Comparison

Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9830	1.0856	0.9281	1.1697	36.3835	-0.2433	16.53%	33.40%	77.91%
+ Geospatial	0.9915	1.1103	0.9087	1.2218	41.0632	-0.3675	14.81%	30.19%	72.56%
% Difference	0.86%	2.27%	-2.09%	4.46%	12.86%	-51.05%	-10.39%	-9.61%	-6.87%

Model VI

Table 11: Model Metrics: LN-TASP Model II Comparison

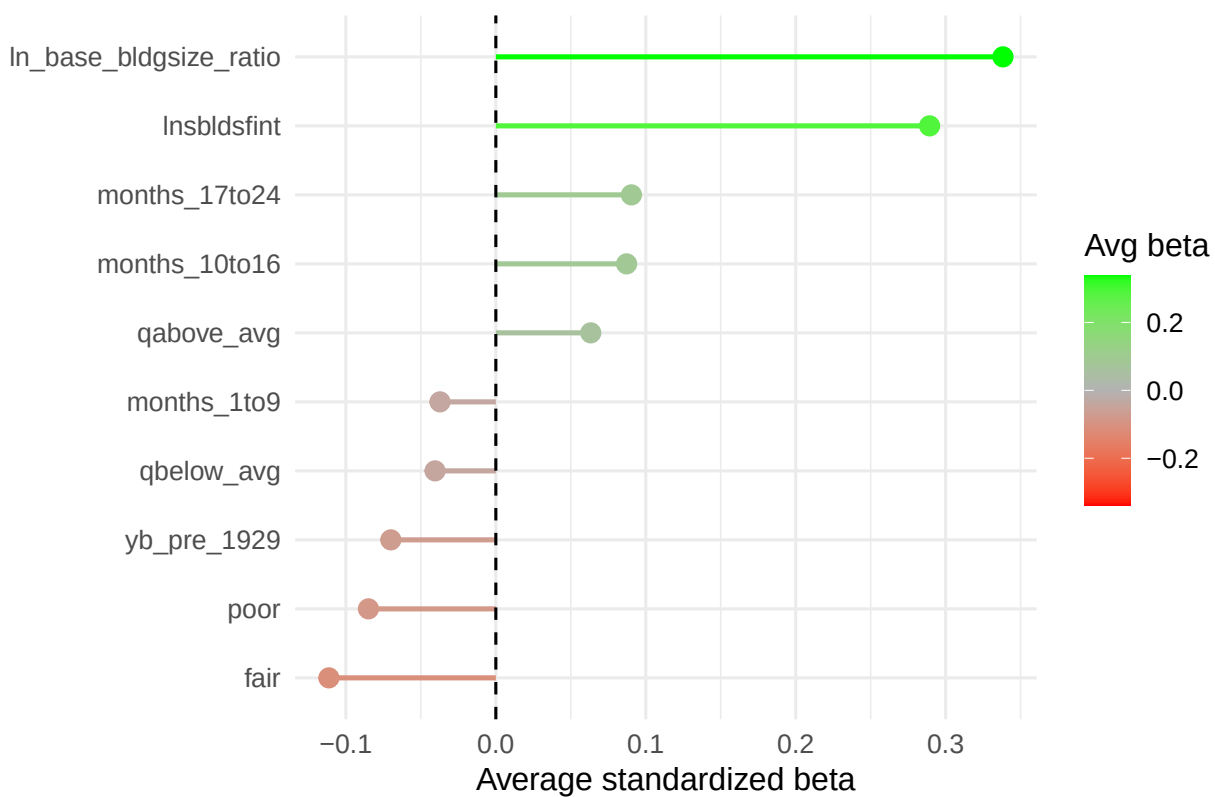
Metric Source	n	df_resid	df_model	r	R^2	Adj. R^2	SEE	f_stat	AIC	BIC
Control	18399	18212	186	0.7533	0.5674	0.5630	0.4075	128.4361	19371	20842
+ Geospatial	18421	18396	24	0.6624	0.4388	0.4380	0.4585	599.2749	23577	23780
% Difference	0.12%	1.01%	-87.10%	-12.06%	-22.67%	-22.19%	12.51%	366.59%	21.71%	14.10%

Table 12: Ratio Metrics: LN-TASP Model II Comparison

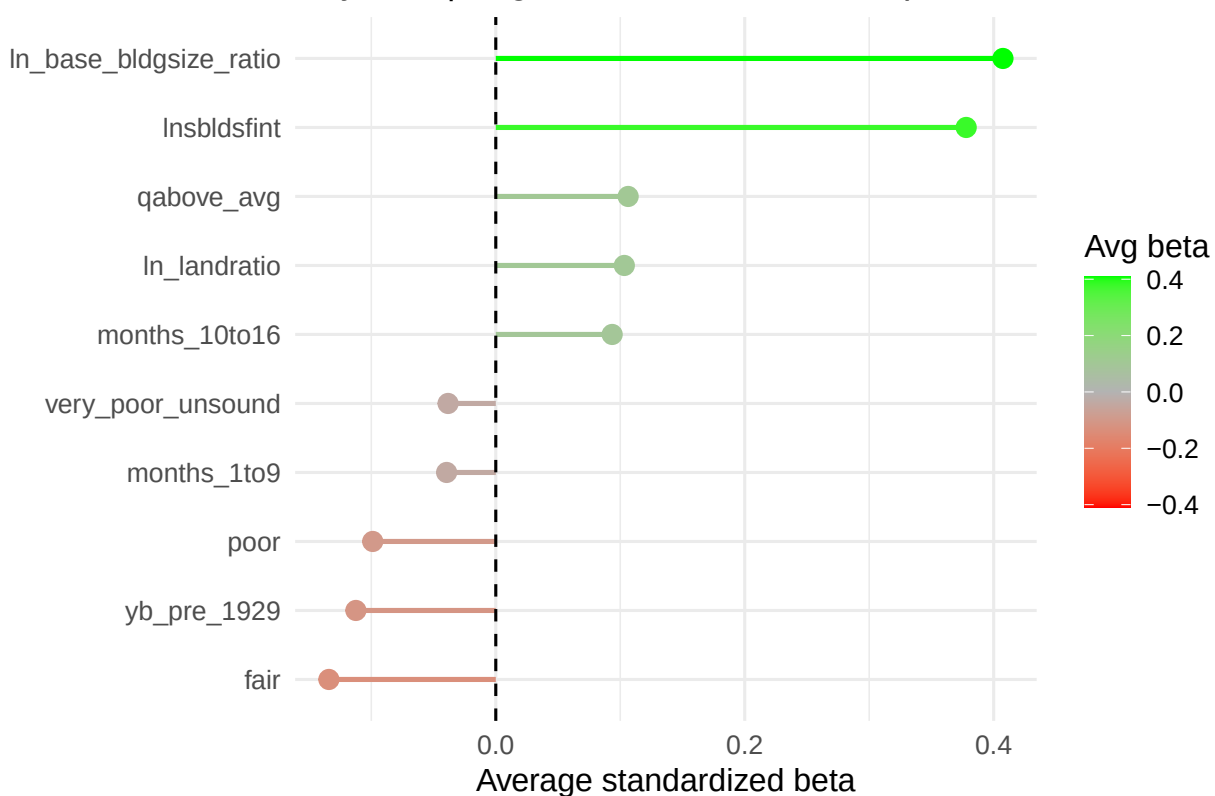
Metric Source	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	Within 10%	Within 20%	Within 50%
Control	0.9835	1.0857	0.9276	1.1705	36.3795	-0.2436	16.34%	33.28%	77.85%
+ Geospatial	0.9933	1.1103	0.9067	1.2246	40.9805	-0.3679	14.90%	30.28%	72.55%
% Difference	0.99%	2.27%	-2.25%	4.62%	12.65%	-51.01%	-8.79%	-9.03%	-6.81%

Model Comparison

Citywide | Avg Std Beta across Initial Models



Citywide | Avg Std Beta across Geospatial Models



Prediction Testing

Table 13: Deviation Summary: Initial LN-SPPSF Model

Metric	Value
Average sale price deviation	-11459.34
Max deviation	508470.20
Min deviation	-1837312.32
Standard deviation	56580.10

Table 14: Deviation Summary: LN-SPPSF Model + Geospatial

Metric	Value
Average sale price deviation	-11404.39
Max deviation	494710.11
Min deviation	-1826999.51
Standard deviation	56055.95